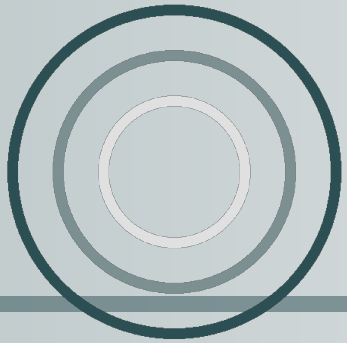
Yanqing An

An Introduction to Legal Intelligence

A first course

First Edition

beautybook



PREFACE

This short book collects the definitions and first theorems of graph colouring, aimed at a reader who has seen basic graph theory but not yet chromatic numbers.

– Grace Hopper

2026

We are on the right scientific path and moving forward. I hope this book will be of help to you.

– Yanqing An

2026

目录

I 法律智能 1

1 概述 3

1.1 Definitions 3

2 法条预测到文书生成 4

2.1 基于神经网络的法条预测 4
2.2 基于SVM的刑事法律判决预测 4
2.3 基于预训练模型的类案检索 4

3 神经网络、SVM与预训练 5

3.1 5

4 大模型时代下的法律智能 6

II 相关技术 7

5 机器学习与神经网络 9

6 SVM 10

7 cnn, rnn, lstm,seq2seq 11

7.1 GAN 11

8 Transfomer 12

8.1 positionencoding 12

8.2 ffn . . . 12
8.3 multiheadattention . . . 12
8.3.1 self attention . . . 12
8.3.2 cross attention . . . 12
8.3.3 casual attention . . . 12
8.4 encoder . . . 12
8.5 decoder . . . 12

9 BERT 13

10 Large Language Models 14

10.1 Definitions . . . 14

III 法律智能技术与应用 **15**

11 现状 17

12 法律判决预测 18

12.1 简介 . . . 18
12.2 Definitions . . . 18

13 类案检索 19

13.1 简介 . . . 19
13.2 Definitions . . . 19
13.3 Challenges . . . 19
13.4 Structure-based Pre-trained Language Model . . . 20
13.5 Legal Case Retrieval with behaviors . . . 20
13.6 Legal Case Retrieval with agents . . . 20

14 律师画像 21

14.1 Definitions . . . 21

15 庭审观点生成 22

15.1 Definitions . . . 22

16 法律问答 23

16.1 Definitions 23

IV 未来发展 24

17 应用场景 26

17.1 Definitions 26

18 未来方向 27

18.1 Definitions 27

V Foundations 28

19 Proper Colourings 30

19.1 Definitions 30

20 The Four Colour Theorem 31

20.1 Introduction 31

PART I

法律智能

*The first part develops the basic vocabulary of Legal Intelligence.*第一部分介绍法律智能的基本概念。



概述

Part I

Sec 1.1 Definitions



法条预测到文书生成

Part I

Sec 2.1 基于神经网络的法条预测

Sec 2.2 基于SVM的刑事法律判决预测

Sec 2.3 基于预训练模型的类案检索

3

神经网络、SVM与预训练

Part I

Sec 3.1



大模型时代下的法律智能

Part I

Part II

相关技术

The first part develops the basic vocabulary of graph colouring and proves the elementary upper bounds every later result builds on.



机器学习与神经网络

Part II

Sec 7.1 GAN

判别器
生成器

TRANSFOMER

Part II

8

Sec 8.1 positionencoding

Sec 8.2 ffn

Sec 8.3 multiheadattention

8.3.1 self attention

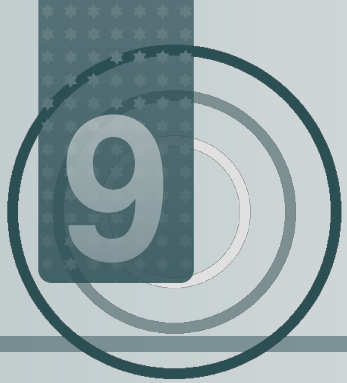
8.3.2 cross attention

8.3.3 casual attention

Sec 8.4 encoder

Sec 8.5 decoder

模拟器



BERT

Part II



LARGE LANGUAGE MODELS

Part II

Sec 10.1 Definitions

Part III

法律智能技术与应用

The first part develops the basic vocabulary of graph colouring and proves the elementary upper bounds every later result builds on.

市面上有大量的法律工具，包括中国裁判文书网[1]、北大法宝[2]等案件检索数据库,以及法小飞[3]、Fabao AI[4]等插件小程序。

中国法律智能技术评测(Challenge of AI in Law, CAIL)[5]从2018年举办至今，提供了部分应用场景。

[1] https://wenshu.court.gov.cn/

[2] https://www.pkulaw.com/

[3] https://sf.xfyun.cn/services/IFlyLegal

[4] https://www.fabao.law/

[5] http://cail.cipsc.org.cn/

法律判决预测

12

Part III

Sec 12.1 简介

刑事、民事、可解释、多被告

Sec 12.2 Definitions

13

类案检索

Part III

Sec 13.1 简介

相似案例匹配、类案检索、对话式类案检索

Sec 13.2 Definitions

Generally speaking, the relevance judgment task in our study is to determine the relevance of the candidate case, given a query case.[1] Legal case retrieval is a specialized IR task aiming to retrieve supporting cases given a query case.[3]

Sec 13.3 Challenges

Long： Both the query and candidate cases involve extreme long texts. For example, cases in COLIEE 2019 Task 1 contain around 3,000 words on average.[5] Legal case documents are typically long text sequences with intrinsic writing logic. Case documents usually consist of five parts: Procedure, Fact, Reasoning, Decision, and Tail.[2] Although pre-trained language models (PLMs) have achieved outstanding performance in various Natural Language Processing (NLP) and IR tasks, they are challenged in legal case retrieval given the length limitation. [7] Retrieval: some shortcomings of general IR in meeting the needs arising from the legal domain.[6] The concept of relevance in the legal domain is quite different from that in general search. Cases without key legal elements or with different key legal elements may lead to different judgments.[2] the target users: legal workers with professional knowledge in law. Compared with web search engines, legal databases (e.g., Westlaw) are preferred considering the data authority and legal effect. The retrieved results are mostly semi-structured case documents rather than web pages. [3] Most existing models retrieve legal cases through lexical or semantic matching between two case documents. However, without domain knowledge, a model cannot understand the relevance in the legal domain comprehensively.[7] Relevance： the definition of legal relevance is beyond topical relevance, involving similar legal issues, facts, consequences, etc.[3] Dataset Collecting a large dataset for this task can be challenging if not impossible. (1)downloading large-scale legal documents is restricted in many law systems. (2)it is quite expensive to obtain accurate relevance judgments since it requires expert knowledge in the legal domain.[5]

Sec 13.4 Structure-based Pre-trained Language Model

SAILER MUSER 预训练模型表征 CaseEncoder: A Knowledge-enhanced Pre-trained Model for Legal Case Encoding. 预训练模型 BERT-PLI: Modeling Paragraph-Level Interactions for Legal Case Retrieval. 数据集 LeCaRD: A Legal Case Retrieval Dataset for Chinese Law System. 结构信息 Incorporating Structural Information into Legal Case Retrieval. 结构信息 SAILER: Structure-aware Pre-trained Language Model for Legal Case Retrieval.

Sec 13.5 Legal Case Retrieval with behaviors

用户决策过程 Investigating the Influence of Legal Case Retrieval Systems on Users' Decision Process. 用户子主题标注 Result Diversification for Legal case Retrieval. 用户意图分类 An Intent Taxonomy of Legal Case Retrieval. 用户判决相关性 Understanding Relevance Judgments in Legal Case Retrieval. 用户行为 Investigating User Behavior in Legal Case Retrieval.

Sec 13.6 Legal Case Retrieval with agents

生成对话问题 Leveraging Event Schema to Ask Clarifying Questions for Conversational Legal Case Retrieval. 对话智能体 Investigating Conversational Agent Action in Legal Case Retrieval. 对话智能体 Query Generation and Buffer Mechanism: Towards a better conversational agent for legal case retrieval. 对话式 Conversational vs Traditional: Comparing Search Behavior and Outcome in Legal Case Retrieval.

Sec 14.1 Definitions

Definition 14.1.1 (Proficiency Assessment)**.** *Given a set of lawyers* $L = \{l_1, l_2, \ldots, l_N\}$*, a set of cases* $C = \{c_1, c_2, \ldots, c_M\}$*, and a set of domains* $F = \{f_1, f_2, \ldots, f_K\}$*, the goal is to identify the proficiencies* $h^l \in (0, 1)^{1\times K}$ *of lawyers in each domain.*

Definition 14.1.1 is the 能力评估 An et al. (2021) *proper* k*-colouring*

庭审观点生成



Part III

Sec 15.1 Definitions

16

法律问答

Part III

Sec 16.1 Definitions

Part IV

未来发展

The first part develops the basic vocabulary of graph colouring and proves the elementary upper bounds every later result builds on.

Sec 17.1 Definitions



未来方向

Part IV

Sec 18.1 Definitions

Part V

Foundations

The first part develops the basic vocabulary of graph colouring and proves the elementary upper bounds every later result builds on.

19 Proper Colourings

Sec 19.1 Definitions

Definition 19.1.1 (Chromatic number)**.** *The* chromatic number $\chi(G)$ *is the least* k *for which* G *admits a proper* k*-colouring.*

Theorem 19.1.1 (Greedy bound)**.** *For any graph* G *with maximum degree* Δ*,* $\chi(G) \leq \Delta + 1$*.*

***Proof*.** Order the vertices arbitrarily and colour them in that order, at each step assigning the least colour not used by an already-coloured neighbour. Each vertex has at most Δ neighbours, so at most Δ colours are forbidden, leaving one of the first $\Delta + 1$ colours free. ∎

Example 19.1 (An odd cycle)**.** *The cycle* C_5 *has* $\Delta = 2$*, so Theorem 19.1.1 gives* $\chi(C_5) \leq 3$*. In fact* $\chi(C_5) = 3$*: two colours are never enough for a cycle of odd length, since the colours would have to alternate and cannot close up consistently.*

Lemma 19.1.1 (Bipartite graphs)**.** *A graph* G *is bipartite if and only if* $\chi(G) \leq 2$*.*

Remark. *Lemma 19.1.1 makes* 2*-colourability and bipartiteness the same question, and both can be checked in linear time by breadth-first search.*

Proposition 20.0.1 (Planar upper bound)**.** *Every planar graph satisfies* $\chi(G) \leq 4$.

Proposition 20.0.1 is the Four Colour Theorem, first proved by computer-assisted case analysis by Appel and Haken (1977), and later given a more tractable proof by Robertson et al. (1997).

Sec 20.1 Introduction

法律智能的现状与发展

Overall, our contributions are summarized below:

-
-
-

参考文献

Yanqing An, Qi Liu, Han Wu, Kai Zhang, Linan Yue, Mingyue Cheng, Hongke Zhao, and Enhong Chen. Lawyerpan: A proficiency assessment network for trial lawyers. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, pages 5–13, 2021.

Kenneth Appel and Wolfgang Haken. Every planar map is four colorable. *Illinois Journal of Mathematics*, 21(3):429–490, 1977.

Neil Robertson, Daniel Sanders, Paul Seymour, and Robin Thomas. The four-colour theorem. *Journal of Combinatorial Theory, Series B*, 70(1):2–44, 1997.